MPCA Project

TITLE OF THE PROJECT: Smart dustbin lid

Team Members:

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Components used:

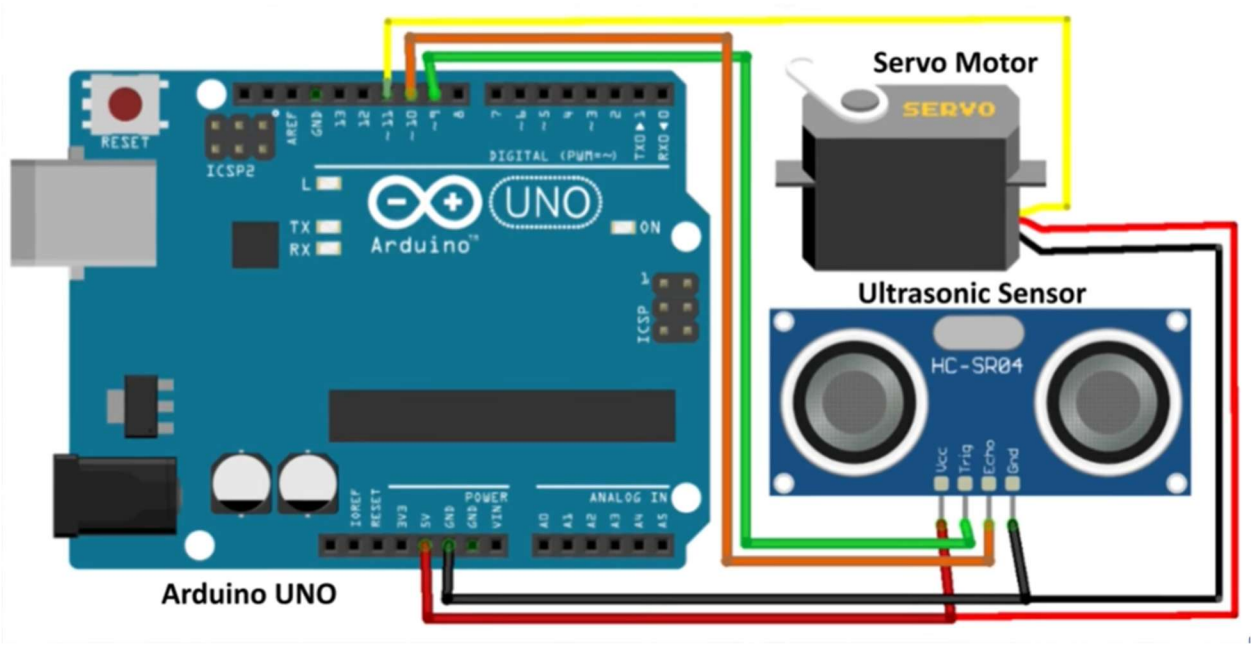
Component	Description	Quantity
Arduino Uno	Microcontroller board based on the ATmega328P.	1
HC-SR04 Sensor	Ultrasonic distance sensor for proximity detection.	1
SG90 Servo	360-degree Continuous Rotation Servo motor.	1
Jumper Wires	Male-to-Male and Male-to-Female connectors.	Set
Power Source	9V Battery or USB Power Supply.	1

Explanation about the project :

This project implements an automated lid-opening mechanism for a dustbin to facilitate a touchless experience, significantly reducing the spread of germs. The system operates on the principle of ultrasonic wave reflection. The **HC-SR04 sensor** constantly monitors the area in front of the bin. When an object comes within a pre-defined threshold (20 cm), the Arduino triggers the **SG90 Servo motor**.

Because this implementation uses a continuous rotation servo, the opening and closing are controlled through precise timing intervals. The motor rotates for a specific duration to lift the lid, pauses for 3 seconds to allow for waste disposal, and then rotates in the opposite direction to securely close the lid.

Circuit Diagram:



Pseudocode:

- **START**
- **Initialize Pins: Trig (9), Echo (8), Servo (10)**
- **WHILE True:**
  - **Trigger Ultrasonic Pulse**
  - **Calculate Distance**
  - **IF Distance $\le 20\text{ cm}$:**
    - **Rotate Servo Forward 170**
    - **Lid Opens**
    - **Wait 3000 ms**
    - **Rotate Servo Backward 190**
    - **Lid Closes**
    - **Stop Servo**
  - **END IF**
  - **Wait (100 ms)**
- **END WHILE**

Source code:

```
#include <Servo.h>

const int trigPin = 9;
const int echoPin = 8;
const int servoPin = 10;

Servo myServo;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
```

```
  myServo.attach(servoPin);
  myServo.write(90); // Neutral Stop signal for continuous servo
}

void loop() {
  long duration;
  int distance;

  // Generate pulse to trigger sensor
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Calculate distance in cm
  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;

  if (distance > 0 && distance <= 20) {
    // 1. OPEN: Rotate for a brief moment to lift the lid
    myServo.write(180);
    delay(170);

    // 2. HOLD: Stop the motor for disposal time
    myServo.write(90);
    delay(3000);

    // 3. CLOSE: Rotate backwards to shut the lid
    myServo.write(0);
    delay(190);
```

```
    // 4. STOP: Reset state
    myServo.write(90);
  }
  delay(100);
}
```

Output Pictures with explanations:

The Smart Dustbin Lid is an automated system designed to provide a hands-free waste disposal experience, improving hygiene and preventing the spread of contaminants. The core functionality of the system relies on ultrasonic distance measurement to detect human presence.